

SHAKYANI JAYASIRIWARDENE

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PhD candidate at the University of Sydney specialising in Human-Computer Interaction and Human-AI Interaction. My research examines how conversational AI personality and communication shape people's expectations, perceptions and behaviour, with applications in digital health and behaviour change. My work spans research ideation, interactive system development and evaluation through laboratory experiments, online studies and longitudinal field studies, combining quantitative and qualitative methods. Alongside nearly four years of professional software engineering experience, I contribute to university teaching, research student mentoring and the coordination of collaborative research projects. I am a quick and enthusiastic learner, keen to explore new research directions and contribute to hands-on projects

Education

University of Sydney

Oct 2023 - present

Doctor of Philosophy, in progress | Expected Thesis submission in Dec 2026

School of Computer Science, Faculty of Engineering

Research focus: Human-Computer Interaction, Human-AI Interaction, Conversational AI, User Experience (UX) Research, AI Personalisation, Ubiquitous Computing, Behaviour Change Technologies, and Digital Health

University of Moratuwa

Sep 2020 - Jan 2023

Master of Science (by Research)

Department of Computer Science, Faculty of Engineering

Thesis: An Adaptive Software Architectural Framework for an Interactive Learning Toolkit

Research focus: Human-Computer Interaction, Software Architecture, Educational Technology, Adaptive and Personalised Learning, Mobile Learning, Recommender Systems

University of Moratuwa

Jan 2016 - May 2020

BSc Honours in Information Technology and Management

Faculty of Information Technology | Gold Medallist, 2020 | Dean's List in 6 of 7 semesters

Research Experience

Doctoral research, University of Sydney

Oct 2023 - present

- **Voice assistant failure mitigation.** Investigated how voice assistant personality and failure-response strategies influence user empathy, satisfaction, and trust in a controlled laboratory study with 60 participants. Published in *ACM IMWUT*, CORE A* (2025).
- **Context-sensitive conversational AI personality.** Investigated how users' expectations of AI personality vary across conversational contexts and evolve during interactions. Developed and deployed a chatbot with eight personality traits adjustable in real time, evaluated through an online mixed-methods study with 60 participants across three contexts. Accepted in *i-com: Journal of Interactive Media, Special Issue on "Generative AI for Interactive Systems,"* guest-edited by Prof. Thomas Kosch (Humboldt University of Berlin) and Prof. Sven Mayer (TU Dortmund University).
- **AI health coaching for physical activity.** Led a 21-day field study in collaboration with Monash University, evaluating an AI voice health coach that encouraged walking around participants' daily schedules. Examined how communication style and message framing influenced motivation for physical activity, with 27 participants completing the study. Manuscript under review at *CHI 2027*.

- **Routine elicitation for just-in-time adaptive interventions (JITAI).** Investigated how users' spoken descriptions of daily routines can be translated into structured representations to inform intervention timing. Conducted a qualitative analysis, examining mismatches between users' intended meaning and AI-generated representations, alongside differences between schedule availability and willingness to engage in physical activity. Developed design recommendations for context-aware conversational health agents. Accepted at *OzCHI 2026*.
- **Inspectable neuro-symbolic architecture for scheduling JITAI.** Investigated a neuro-symbolic architecture combining LLM-based information extraction, a persistent knowledge graph and finite-state-machine dialogue management to make routine elicitation and intervention planning traceable. Evaluated the perceived accuracy of generated schedule summaries in a laboratory feasibility study with 16 participants. Short paper accepted for *UbiComp 2026 Workshop: XAIforU, Explainable AI for Ubiquitous, Pervasive and Wearable Computing*.
- Conducted HCI research across the full research lifecycle, from ideation, study design and prototype development to conducting laboratory experiments, online studies and longitudinal field studies, analysing data, identifying research and design implications, and writing manuscripts. Coordinated research teams and collaborated with colleagues, applying statistical modelling in R and qualitative coding to analyse survey, behavioural, interview and focus-group data.

Master's research, University of Moratuwa

Sep 2020 - Jan 2023

- Developed iLearn, an adaptive and interactive mobile learning toolkit, using Android Studio, Java, SQLite and Firebase.
- Investigated software architecture and knowledge-based algorithms for recommending interactive learning assessments.

Publications

Journal articles

Jayasiriwardene, Shakyani, Zhou, Hongyu, Jiang, Weiwei, Tag, Benjamin, Koemel, Nicholas A., Ahmadi, Matthew, Goncalves, Jorge, Stamatakis, Emmanuel, Withana, Anusha and Sarsenbayeva, Zhanna. "Probing context-sensitive expectations through user-driven personality configuration in conversational AI" *i-com*, 2026. <https://doi.org/10.1515/icom-2026-0029>

Jayasiriwardene, S., Tag, B., Withana, A., & Sarsenbayeva, Z. (2025). More Than Words: The Impact of Voice Assistant Personality Traits on Failure Mitigation. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, 9(3), Article 91, 33 pages. doi.org/10.1145/3749518

Jayasiriwardene, S., & Meedeniya, D. (2023). An adaptive and interactive learning toolkit (iLearn). *Software Impacts*, 15, 100471. doi.org/10.1016/j.simpa.2023.100471

Conference papers

Jayasiriwardene, S.*, Mountford, B.*, Ma, M., van Berkel, N., Goncalves, J., Koemel, N., Ahmadi, M., Stamatakis, E., & Sarsenbayeva, Z. (2026). Graphing the Everyday: A Neurosymbolic Approach to Eliciting Routines for Just-In-Time Adaptive Interventions. *OzCHI 2026*. **Accepted.**

Jayasiriwardene, S., & Meedeniya, D. (2022). A Knowledge-based Adaptive Algorithm to Recommend Interactive Learning Assessments. *2nd International Conference on Advanced Research in Computing (ICARC)*, pp. 379-384. doi.org/10.1109/ICARC54489.2022.9753913

Jayasiriwardene, S., & Meedeniya, D. (2021). Interactive and Adaptive Learning Content Authoring Framework for an M-Learning Toolkit. *1st Conference on Online Teaching for Mobile Education (OT4ME)*, pp. 153-160. doi.org/10.1109/OT4ME53559.2021.9638890

Jayasiriwardene, S., & Meedeniya, D. (2021). Architectural framework for an interactive learning toolkit. *International Research Conference on Smart Computing and Systems Engineering (SCSE)*, pp. 14-21. doi.org/10.1109/SCSE53661.2021.9568330

Workshop and doctoral colloquium papers

Jayasiriwardene, S. (2026). The Impact of Conversational AI Personality on Human Perceptions and Behaviour. *UbiComp/ISWC 2026 Doctoral Colloquium*. **Accepted.**

Jayasiriwardene, S.*, Mountford, B.*, Ma, M., Jiang, W., van Berkel, N., Koemel, N., Ahmadi, M., Goncalves, J., Stamatakis, E., Withana, A., & Sarsenbayeva, Z. (2026). An Inspectable Neuro-Symbolic Pipeline for Routine Elicitation in Conversational Health Coaching. *XAI for U 2026: 3rd International Workshop on Explainable AI for Ubiquitous, Pervasive and Wearable Computing, UbiComp/ISWC 2026*. **Accepted short paper.**

Manuscripts under review

Jayasiriwardene, S., Ma, M., Kummerfeld, J. N., Koemel, N., Ahmadi, M., Stamatakis, E., van Berkel, N., Withana, A., & Sarsenbayeva, Z. (2026). Effect of an AI Coach's Communication Style and Message Framing on Motivating Physical Activity. **Under review.**

Teaching and Supervision

Casual Academic, University of Sydney

Feb 2024 - present

- Deliver tutorials in Human-Computer Interaction and Usability Engineering to classes of up to 35 students.
- Mark student assessments and contribute to regular teaching-team planning meetings and examination preparation.

Honours and research intern mentoring, University of Sydney

2024 - present

- Contribute to the research supervision and mentoring of four Honours students and two research interns, supporting their research progress and development as independent researchers.

Visiting Instructor, University of Moratuwa

Feb 2023 - May 2023

- Conducted Big Data Analytics tutorials for Honours students at the Faculty of Information Technology.

Temporary Assistant Teacher, Musaeus College

Aug 2015 - Dec 2015

- Taught Information Technology for Advanced Level students.

Industry Experience

IFS R&D International, Colombo, Sri Lanka

Mar 2020 - Oct 2023

Senior Software Engineer/ Software Engineer, Extensibility Team

- Developed and maintained backend components of the IFS Applications ERP platform, supporting new and existing product versions.
- Contributed to application extensibility through development involving IFS Aurena, REST APIs and OData protocol.
- Participated across the software development lifecycle, including requirements analysis, software design, implementation, debugging, test-case development and developer testing.
- Maintained source code in GitLab, and mentored interns and newly appointed software engineers.

Software and tools: Java, SQL, Oracle PL/SQL, JavaScript, Angular, IFS Aurena, IFS Developer Studio, REST APIs, OData and GitLab.

IFS R&D International, Colombo, Sri Lanka

Aug 2018 - Feb 2019

Software Engineering Intern, Developer Studio Team

- Contributed to a research project developing a language server for Marble, IFS's domain-specific language, using the Language Server Protocol (LSP) and integrating it with Eclipse Che open source project.
- Developed custom tooling features using Eclipse Che's extension capabilities.

Software and tools: Java, Xtext, Xtend, Eclipse IDE, Eclipse Che, Docker, Maven, Gradle and ANTLR4.

Awards and Scholarships

- Faculty of Engineering winner, and University of Sydney finalist, 3-Minute Thesis competition, 2026.
- Poster Award (Second Place), PRESS HDR Student Conference, University of Sydney, 2025.
- “Beyond” Award for excellent contribution as a student volunteer, UbiComp/ISWC 2025.
- University of Sydney International Tuition Fee Scholarship, awarded for the full PhD candidature.
- Faculty of Engineering Research Stipend Scholarship, awarded for the full PhD candidature.
- Best Full Paper Award, 1st Conference on Online Teaching for Mobile Education (OT4ME), 2021.
- Finalist, VidyaJyothi Professor Dayantha S Wijeyesekera Award for the Most Outstanding Graduated of the Year, University of Moratuwa, 2020.

Research and Travel Support

- Postgraduate Research Support Scheme grant, University of Sydney, 2026: A\$1,120.
- Postgraduate Research Support Scheme grant and HDR Research Support Funding, University of Sydney, 2025: A\$5,165 in total.

Academic Service and Leadership

Peer review

- MobileHCI 2026, CHI 2026, CHI 2026 Posters, CHI 2025 Student Research Competition, ACM Interactive Surfaces and Spaces (ISS) 2024, MERCon 2025, and OT4ME 2021.

Leadership and mentoring

- Vice President, Postgraduate Research Engineering Students' Society (PRESS), Faculty of Engineering, University of Sydney, 2026/27.
- Committee Member, HDR Student Committee, School of Computer Science, University of Sydney, 2025.
- Senior Buddy, Women in Engineering Network, University of Sydney, 2025.
- Peer Mentor, PG Connect, Faculty of Engineering, University of Sydney, 2025.
- HDR Student Life Grant, University of Sydney, 2025: A\$1,000 for organising a student team-building activity.
- Committee Member, IEEE Women in Engineering Sri Lanka Section, 2023/24.
- Secretary, University of Moratuwa Toastmasters Club, 2022/23.
- Chair, IEEE Women in Engineering Student Branch Affinity Group, University of Moratuwa, 2019/20.
- Editor-in-Chief, FIT Chronicle, Faculty of Information Technology, University of Moratuwa, 2019/20.

Conference service

- Student Volunteer, UbiComp/ISWC, Espoo, Finland, 2025.
- Sub-Committee Member, PRESS HDR Student Conference, University of Sydney, 2025.
- Local Student Volunteer, IEEE International Symposium on Mixed and Augmented Reality (ISMAR), University of New South Wales, 2023.
- IEEE Women in Engineering (Sri-Lanka section) events: WIESymp logistics committee (2021), WIESymp20 publications committee (2020), WIE International Leadership Summit program committee (2021), and SLYWC20 WIE Track session co-host (2021).

Referees

Provided upon request.